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Research Article

The RNA-binding protein MdHYL1 modulates cold tolerance and disease resistance in apple

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Abstract

Apple (*Malus domestica*) trees often experience various abiotic and biotic stresses. However, due to the long juvenile period of apple and its high degree of genetic heterozygosity, only limited progress has been made in developing cold-hardy and disease-resistant cultivars through traditional approaches. Numerous studies reveal that biotechnology is a feasible approach to improve stress tolerance in woody perennial plants. HYPONASTIC LEAVES1 (HYL1), a double-stranded RNA-binding protein, is a key regulator involved in apple drought stress response. However, whether HYL1 participates in apple cold response and pathogen resistance remains unknown. In this study, we revealed that MdHYL1 plays a positive role in cold tolerance and pathogen resistance in apple. MdHYL1 acted upstream to positively regulate freezing tolerance and *Alternaria alternata* resistance by positively modulating transcripts of *MdMYB88* and *MdMYB124* in response to cold stress or *A. alternata* infection. In addition, MdHYL1 regulated the biogenesis of several miRNAs responsive to cold and *A. alternata* infection in apple. Furthermore, we identified Mdm-miRNA156 (Mdm-miR156) as a negative regulator of cold tolerance and Mdm-miRNA172 (Mdm-miR172) as a positive regulator of cold tolerance, and that Mdm-miRNA160 (Mdm-miR160) decreased plant resistance to infection by *A. alternata*. In summary, we highlight the molecular role of MdHYL1 regarding cold tolerance and *A. alternata* infection resistance, thereby providing candidate genes for breeding apple with freezing tolerance and *A. alternata* resistance using biotechnology.

Introduction

MicroRNAs (miRNAs) are a class of endogenous noncoding small RNAs, ranging from 20 to 24 nucleotides in length. In general, miRNAs reduce the expression level of their target genes via repression of transcription and RNA cleavage (Liang et al. 2022). The biogenesis of miRNAs involves several distinct steps (Moran et al. 2017). In *Arabidopsis* (*Arabidopsis thaliana*), a single RNase type III named DICER-LIKE1 (DCL1) and a double-stranded RNA-binding protein known as HYPONASTIC LEAVES1 (HYL1) are required for the processing of pri-miRNAs into pre-miRNA and pre-miRNA into mature miRNA/miRNA*. The entire process takes place in the nucleus with the help of several microprocessor accessory proteins, including SERRATE (SE) and DAWDLE (DDL) (Rogers and Chen 2013; Ledda et al. 2020).

miRNAs are known to be involved in abiotic stress tolerance (Song et al. 2019). The miR156-SPLs module has been found to contribute to heat stress memory in *Arabidopsis* (Stief et al. 2014), and miR156p positively regulates drought tolerance in apple (Niu et al. 2019). In rice (*Oryza sativa*) and *Arabidopsis*, reducing miR165/166 increases drought tolerance by increasing abscisic acid (ABA) content (Yan et al. 2016; Zhang et al. 2018). In addition, apple miR160 positively regulates drought tolerance by facilitating root development and targeting MdARF17 (Auxin Response Factor 17) (Shen et al. 2022). It is also known that cold stress induces miR319, thereby leading to the downregulation of

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TEOSINTE BRANCHED1 and *CYCLOIDEA* genes, which positively regulates cold tolerance in rice and sugarcane (*Saccharum officinarum*) (Thiebaut et al. 2012; Yang et al. 2013). Similarly, cold-inducible miR393 targets the auxin receptor gene TIR/AFB to enhance cold tolerance in switchgrass (*Panicum virgatum* L.) (Liu et al. 2017), and the overexpression (OE) of miR156 decreases cold tolerance by targeting *OsSPL3* in rice (Zhou and Tang 2019).

miRNAs have also been found to play a role in plant immunity (Song et al. 2019). In *Arabidopsis*, miR160a targets ARFs and thereby contributes to PAMP-triggered immunity (PTI) (Li et al. 2010). In rice, miR398b is induced by *Magnaporthe oryzae* infection and enhances disease resistance via PTI (Li et al. 2014). In contrast, miR164a is repressed by *M. oryzae* and induces expression of *OsNAC60* as part of the PTI system (Wang et al. 2018). Finally, it is also known that decreased miR6019/6020 is associated with increased expression of NB-LRRs, which enhances the resistance of mature rice plants to tobacco mosaic virus (Deng et al. 2018).

In plant cells, the dsRNA-binding protein HYL1, the RNase III DCL1, and the C2H2 Zn-finger protein SE are required for the accurate processing of miRNA precursors (Dong et al. 2008). Because of the vital role of HYL1, the *hyl1* mutant plants displayed a series of phenotypes such as shorter stature, delayed flowering, leaf hyponasty, reduced fertility, decreased rate of root growth, and altered root gravitropic response (Vazquez et al. 2004). The *hyl1* mutant also exhibits less sensitivity to auxin and cytokinin and hypersensitivity to ABA (Lu and Fedoroff 2000; Vazquez et al. 2004). In addition, *hyl1* mutants showed misplaced cuts and reduced the biosynthesis of many miRNAs (Sunkar and Zhu 2004; Liu et al. 2008). In response to drought stress, *MdHYL1* induces drought tolerance by regulating the biogenesis of drought responsive Mdm-miRNAs (Khan et al. 2011; Shen et al. 2022). Under both control and drought conditions, *MdHYL1* has been found to be a direct target gene of MdARF17, a negative regulator of drought tolerance (Shen et al. 2022). In addition, MdHYL1 has also been found to interact with MdMYB88 and MdMYB124, 2 positive regulators of cold tolerance and resistance to infection by *Alternaria alternata*. The R2R3 MYB transcription factor MdMYB88 and its paralog MdMYB124 are 2 positive regulators for apple freezing, drought, and pathogen attack (Geng et al. 2018, 2020; Xie et al. 2018). To be specific, MdMYB88 and MdMYB124 positively regulate cold hardiness by directly targeting CBF-independent genes including *COLD SHOCK DOMAIN PROTEIN 3* (*MdCSP3*) as well as CBF-dependent genes including *CIRCADIAN CLOCK ASSOCIATED 1* (*MdCCA1*) (Xie et al. 2018). Moreover, MdMYB88 and MdMYB124 promote anthocyanin accumulation and H_2O_2 detoxification in response to cold. In addition, MdMYB88 and MdMYB124 contribute to pathogen defense by directly and positively regulating the expression of *MdCM2* and the accumulation of phenylpropanoid of apple (Geng et al. 2020). Although MdHYL1 can interact with MdMYB88 and MdMYB124, their genetic and regulatory relationship in response to cold stress and *A. alternata* infection is largely unknown.

In this study, we reveal that MdHYL1 functions to positively regulate cold tolerance and disease resistance by positively regulating expression of *MdMYB88* and *MdMYB124*, as well as by inducing miRNA biogenesis. MdHYL1 responds to cold stress by enhancing cold tolerance; it does so by positively regulating the expression of cold-responsive genes, anthocyanin accumulation, and reactive oxygen species (ROS) detoxification and by regulating the expression of *MdMYB88* and *MdMYB124*. Moreover, after pathogen inoculation, we also found that MdHYL1 induces the accumulation of phenylpropanoids via MdMYB88 and MdMYB124, thereby increasing disease resistance. Finally, we also found that MdHYL1 affects the biogenesis of miRNAs, which in turn play important roles in the cold tolerance and defense response of apple.

Results

MdHYL1 enhances the expression of *MdMYB88* and *MdMYB124* under cold stress

In a previous study, we found that MdHYL1 interacts with 2 MYB transcription factors, MdMYB88 and MdMYB124, which positively regulate cold tolerance (Xie et al. 2018; Niu et al. 2022). This suggests that MdHYL1 may play some role in the apple cold stress response.

To explore the function of *MdHYL1* under cold stress, we determined its expression levels in apple (*Malus domestica*) cv. 'Golden Delicious' treated with low temperature (4 °C) relative to a control treatment. We found that *MdHYL1* was slightly induced by low-temperature treatment (Supplemental Fig. S1A), further indicating that *MdHYL1* may play a role in cold stress tolerance. Despite the interaction between MdHYL1 and MdMYB88 or MdMYB124, the expression of *MdHYL1* was not affected in both *MdMYB88* and *MdMYB124* OE plants and *MdMYB88/124* RNA interference (RNAi) transgenic lines compared to GL-3 under cold, and neither MdMYB88 nor MdMYB124 was able to bind to the promoter region of *MdHYL1* (Supplemental Fig. S1, B to D). However, the expression of both *MdMYB88* and *MdMYB124* decreased in *MdHYL1* RNAi transgenic plants and increased in *MdHYL1* OE transgenic plants (Supplemental Fig. S1, E to H). Moreover, we detected the protein level of MdMYB88 and MdMYB124 in *MdHYL1* RNAi transgenic plants in response to cold treatment. The results showed that MdHYL1 positively regulated the protein accumulation of MdMYB88 and MdMYB124 under cold, consistent with the regulation of *MdMYB88* and *MdMYB124* by MdHYL1 at the mRNA level (Supplemental Fig. S1I). Taken together, these results suggest that MdHYL1 positively regulates the expression of *MdMYB88* and *MdMYB124* under cold stress, but not vice versa.

MdMYB88 and MdMYB124 can regulate the expression of cold-responsive genes, including *MdCBF1*, *MdCBF3*, *MdCOR47*, *MdCCA1*, and *MdCSP3*. Of these genes, both

MdCCA1 and *MdCSP3* are direct targets of MdMYB88 and MdMYB124 (Xie et al. 2018). We were then interested whether MdHYL1 regulated the expression of these cold-responsive genes during the apple cold response. Our reverse transcription quantitative PCR (RT-qPCR) results showed significant repression of *MdCBF1*, *MdCBF3*, *MdCOR47*, *MdCCA1*, *MdCSP3*, and *MdSIZ1* in *MdHYL1* RNAi transgenic plants exposed to cold conditions, relative to the expression levels of GL-3 plants (Fig. 1). Moreover, we also found that the expression of these cold-response genes was elevated in *MdHYL1* OE transgenic plants subjected to cold treatment (Fig. 1), which suggests that they are positively regulated by MdHYL1.

MdCBF1, *MdCBF3*, *MdSIZ1*, *MdCCA1*, and *MdCOR47* are well-known cold marker genes in plant cold response (Jaglo-Ottosen et al. 1998; Miura et al. 2007; Kim et al. 2009; Dong et al. 2011; Yang et al. 2022). CSP3 in *Arabidopsis* plays a positive role in cold tolerance (Kim et al. 2009). To understand the role of its counterpart in apple, we subjected transgenic plants OE *MdCSP3* to freezing stress (Supplemental Fig. S2, A and B) (Li et al. 2021). As expected, after treatment, *MdCSP3* OE transgenic lines displayed increased survival rates and decreased levels of ion leakage relative to GL-3 plants (Supplemental Fig. S2, C and D). This suggests that *MdCSP3* plays a positive role in cold tolerance.

MdHYL1 positively regulates cold tolerance in apple

Regulation of *MdMYB88* and *MdMYB124* and their target genes by MdHYL1 under cold prompted us to investigate the role of MdHYL1 in apple cold tolerance. To explore the biological function of MdHYL1 in the apple cold response, we evaluated the cold tolerance of *MdHYL1* RNAi transgenic plants that we generated in a previous study (Shen et al. 2022). Without cold acclimation, the survival rate of *MdHYL1* RNAi transgenic plants (30% to 33%) was lower than that of nontransgenic plants (GL-3 plants, 48%) after freezing treatment (Fig. 2, A and B). After cold acclimation, though the survival rates of both GL-3 and *MdHYL1* RNAi transgenic plants were higher, we found that *MdHYL1* RNAi transgenic plants (38% to 41% survival rate) were still more sensitive to freezing stress than GL-3 plants (58% survival rate) (Fig. 2, C and D). Regardless of acclimation treatment, the ion leakage rate of *MdHYL1* RNAi transgenic leaves (82% to 85%) was higher than GL-3 leaves (67%) (Fig. 2E). In addition, the dormant branches taken from 1-yr-old *MdHYL1* RNAi transgenic plants also showed higher ion leakage rate (74%) than GL-3 plants (58%) (Fig. 2F). These results suggest that *MdHYL1* RNAi transgenic plants were more sensitive to freezing stress than GL-3 plants.

We also examined the cold tolerance of *MdHYL1* OE transgenic plants, which we had also generated in a previous study (Shen et al. 2022). In contrast to *MdHYL1* RNAi transgenic plants, *MdHYL1* OE transgenic plants were more tolerant to freezing stress, with a higher survival rate (61% to 63%) than GL-3 plants (49%) before cold acclimation (Fig. 3, A and B). Cold acclimation increased the survival rates of both GL-3 and *MdHYL1* OE transgenic plants; however, *MdHYL1* OE transgenic plants still had a higher survival rate (78%) than GL-3 plants (58%) (Fig. 3, C and D). Moreover, ion leakage in the leaves of *MdHYL1* OE transgenic plants (34% to 49%) was lower than that of GL-3 plants (43% to 65%) under freezing stress before and after cold acclimation (Fig. 3E). In addition, ion leakage of the dormant branches from 1-yr-old *MdHYL1* OE plants (45% to 47%) was lower than that of GL-3 (63%) when exposed to freezing stress (Fig. 3F).

ROS, which include hydrogen peroxide (H_2O_2) and superoxide anion (O^{2-}), are toxic to plant cells and can be quickly generated during abiotic stress (Yusuf et al. 2008). We found that the *MdHYL1* RNAi transgenic plants produced more H_2O_2 than GL-3 plants after being exposed to low-temperature (4 °C) environments for 8 h (Supplemental Fig. S3A), suggesting that the *MdHYL1* RNAi transgenic plants were more severely damaged by cold than GL-3 plants. To cope with abiotic stresses, including cold stress, plants increase their antioxidant content including anthocyanin and enhance the activity of antioxidant enzymes (Nayyar and Chander 2004). In most plants, peroxidase (POD) and catalase (CAT) are the most important enzymatic antioxidants (Gill and Tuteja 2010). We consistently observed that POD and CAT activities were lower in the *MdHYL1* RNAi transgenic plants than in GL-3 plants (Supplemental Fig. S3, B and C), which suggests that *MdHYL1* RNAi transgenic plants had a weaker ability to scavenge H_2O_2 by hydroxylation under cold than GL-3 plants. In addition, *MdHYL1* OE transgenic plants had higher POD and CAT activities, which led to lower H_2O_2 content in *MdHYL1* OE transgenic plants than GL-3 plants under cold conditions (Supplemental Fig. S4, A to C). Moreover, *MdHYL1* RNAi plants accumulated less anthocyanin, and *MdHYL1* OE plants accumulated more anthocyanin relative to GL-3 control plants (Supplemental Fig. S5). These results indicate that MdHYL1 facilitates apple cold tolerance by accumulating antioxidant enzymes and nonenzymatic antioxidants including anthocyanin.

MdHYL1 acts upstream of *MdMYB88* in response to cold

MdHYL1 interacts with MdMYB88 and its paralog MdMYB124 (Niu et al. 2022). To detect possible genetic interaction between MdHYL1 and MdMYB88, we generated transgenic calli that showed decreased expression of *MdHYL1* (*MdHYL1* RNAi) (Supplemental Fig. S6A) or increased expression of *MdMYB88* (*MdMYB88* OE) (Supplemental Fig. S6B). In addition, we also overexpressed *MdMYB88* in transgenic *MdHYL1* RNAi calli. We then applied a low-temperature treatment to these transgenic calli for 15 d. After cold treatment, *MdHYL1* RNAi calli showed a lower relative growth rate than the wild type, while the relative growth rate of *MdMYB88* OE calli was higher than the wild type (Supplemental Fig. S6, C and D). These findings are consistent with previous research which found that both MdMYB88

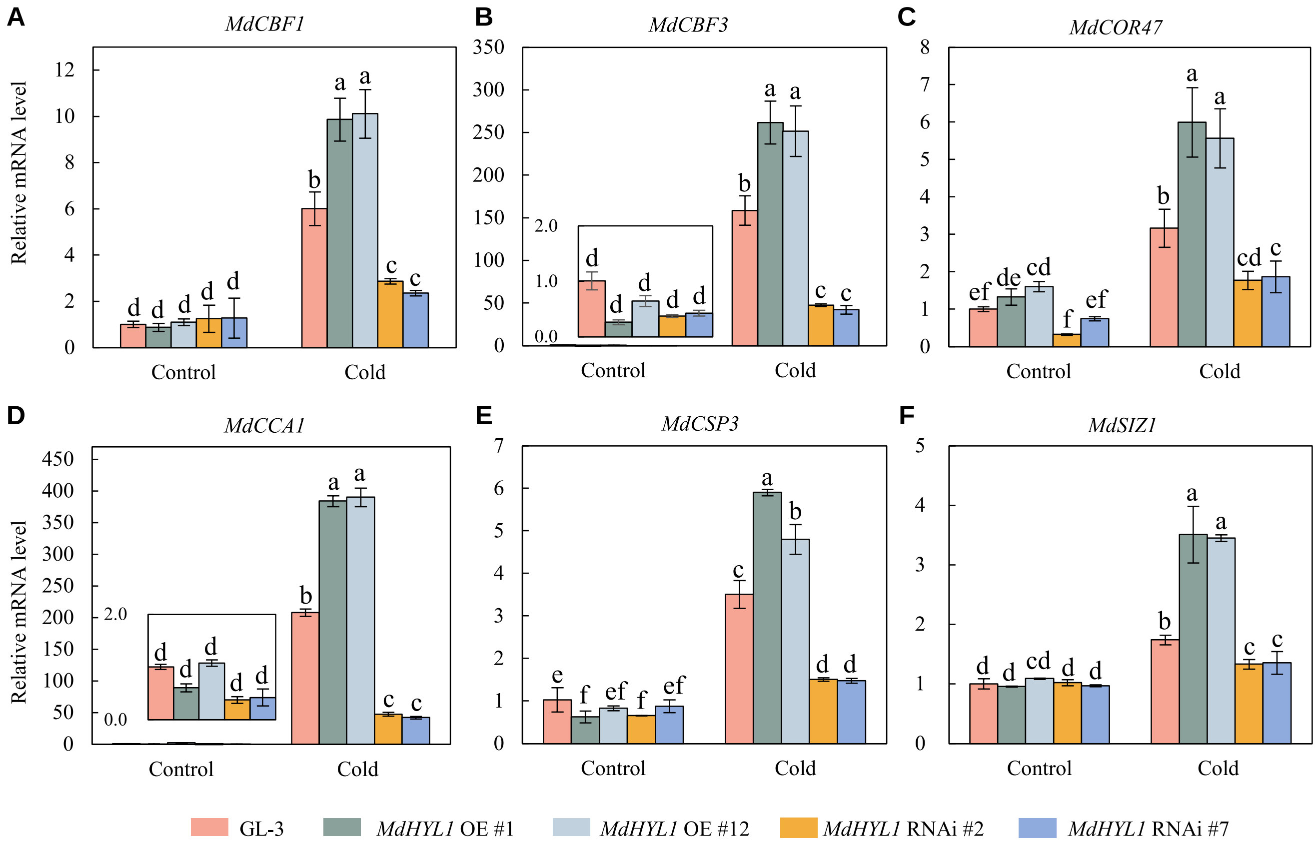

Figure 1. The expression level of cold-responsive genes in GL-3, transgenic *MdHYL1* OE, and RNAi plants under control and cold conditions. **A to F)** Expression levels of *MdCBF1*, *MdCBF3*, *MdCOR47*, *MdCCA1*, *MdCSP3*, and *MdSIZ1* in GL-3, *MdHYL1* OE, and RNAi plants. Plants were treated at 4 °C for 8 h, and leaves were collected for total RNA extraction and RT-qPCR analysis. OE, overexpression. Error bars represent standard deviation ($n = 3$). The letters indicate significant differences according to 1-way ANOVA (Tukey's test; $P < 0.05$).

and MdMYB124 play positive roles with respect to cold tolerance (Xie et al. 2018). Moreover, OE of *MdMYB88* was able to rescue the cold tolerance of *MdHYL1* RNAi transgenic calli, suggesting that MdMYB88 acts downstream of MdHYL1 during the cold response.

MdHYL1 modulates the biogenesis of cold-responsive miRNAs in response to cold

HYL1 has been reported to play a pivotal role in miRNA biogenesis (Tripathi et al. 2022). We then performed sRNA-seq analysis to determine whether *MdHYL1* also regulated the biogenesis of miRNAs in response to cold stress. Pearson correlation analysis showed a good correlation and repeatability between samples (Supplemental Fig. S7). sRNA-seq analysis showed that the biogenesis of 25 Mdm-miRNAs was altered in GL-3 plants after cold treatment, which indicated that apple may respond to cold by altering the biogenesis of cold-responsive Mdm-miRNAs. Relative to GL-3 plants, the accumulation of 96 Mdm-miRNAs was substantially altered in *MdHYL1* RNAi plants exposed to cold stress, including 8 cold-responsive Mdm-miRNAs Moreover, among the affected Mdm-miRNAs, the repression of *MdHYL1* increased the accumulation of 41 Mdm-miRNAs and decreased the accumulation of 55 Mdm-miRNAs, which indicates that MdHYL1 participates in the biogenesis of cold-responsive Mdm-miRNAs in response to cold stress (Fig. 4, A and B). The 8 cold-responsive Mdm-miRNAs included Mdm-miR396-p5, Mdm-miR160, Mdm-miR319-p5, Mdm-miR156, Mdm-miR172, Mdm-miR164e-p3, Mdm-miR159-p3, and Mdm-miR398-p5. Stem-loop RT-qPCR results showed that accumulation of all these 8 Mdm-miRNAs was lower in *MdHYL1* RNAi transgenic plants compared to GL-3 under control conditions. When exposed to cold treatment, the relative transcripts of Mdm-miR159-p3, Mdm-miR172, Mdm-miR319-p5, and Mdm-miR396-p5 were induced by cold in both the GL-3 and *MdHYL1* RNAi plants; however, their abundance was lower in *MdHYL1* RNAi transgenic plants than in GL-3. In contrast, the relative transcripts of Mdm-miR156, Mdm-miR160, Mdm-miR164e-p3, and Mdm-miR398-P5 were repressed by cold in both the GL-3 and *MdHYL1* RNAi plants; however, their abundance was higher in *MdHYL1* RNAi transgenic plants than in GL-3 after cold treatment (Fig. 4C). Taken together, our results suggest that MdHYL1 regulates the biogenesis of cold-responsive miRNAs in response to cold.

To confirm whether these cold-responsive Mdm-miRNAs contribute to cold tolerance in apple, we examined the biological functions of Mdm-miR156 and Mdm-miR172 under cold. In

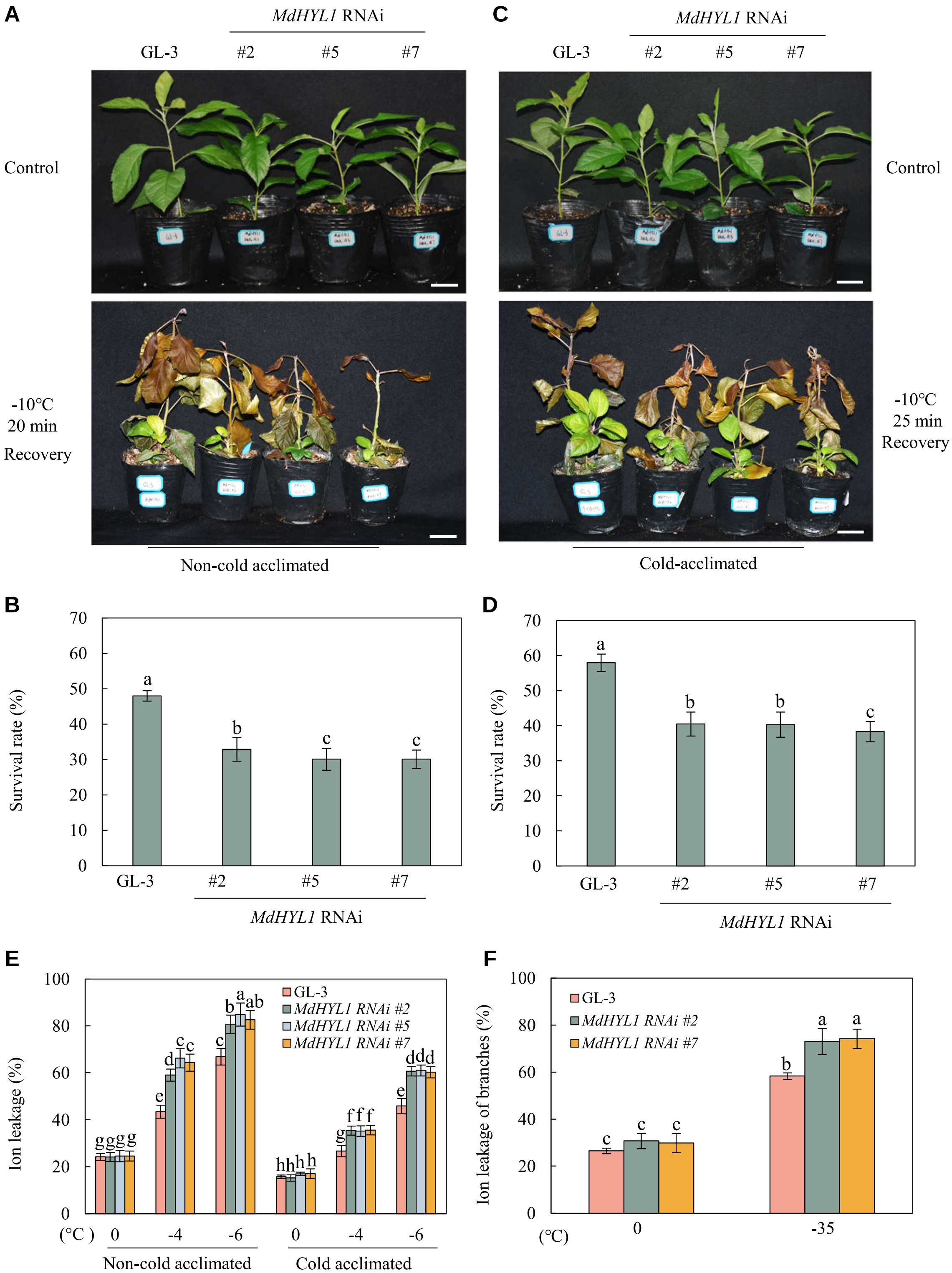

Figure 2. Freezing tolerance of *MdHYL1* RNAi plants. **A** and **C)** Freezing tolerance of *MdHYL1* RNAi plants. One-month-old plants were transferred to soil for an additional 1 mo, and GL-3 and *MdHYL1* RNAi plants without or with cold acclimation were exposed to −10 °C for 20 min and −10 °C for 25 min, respectively. After freezing treatment, all plants were transferred to a growth chamber for recovery. Bars = 3 cm. **B** and **D)** Survival rate of the plants shown in **A** and **C**. **E)** Ion leakage of noncold-acclimated and cold-acclimated GL-3 and *MdHYL1* RNAi plants. One-month-old plants were transferred to soil for an additional 1 mo, and leaves were collected for an ion leakage assay. **F)** Ion leakage from dormant branches. Error bars indicate standard deviation ($n = 12$–15 in **B** and **D** and 6 in **E** and **F**). The letters indicate significant differences according to 1-way ANOVA (Tukey's test; $P < 0.05$).

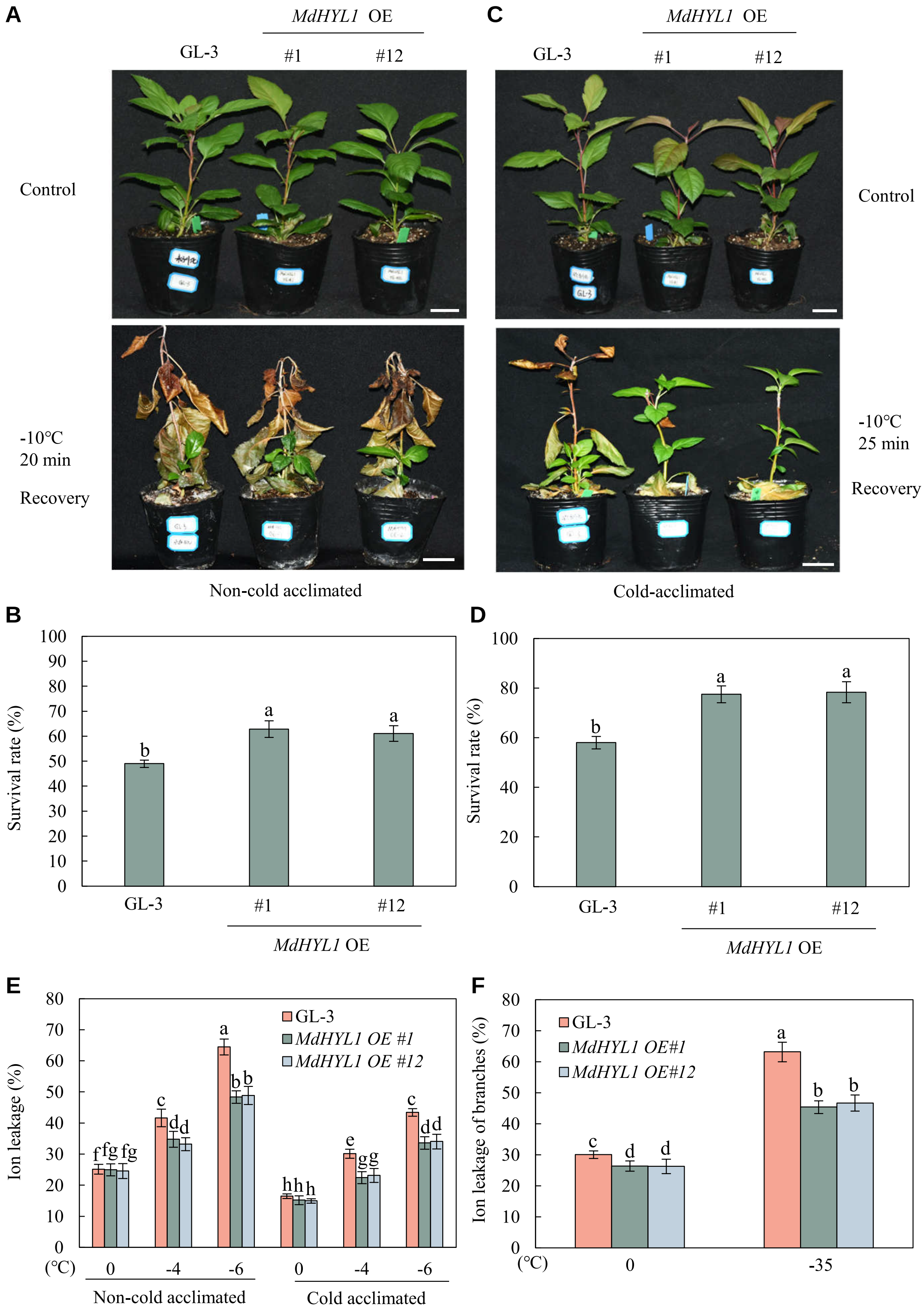

Figure 3. Freezing tolerance of *MdHYL1* overexpression (OE) plants. **A** and **C)** Freezing tolerance of *MdHYL1* OE plants. One-month-old plants were transferred to soil for an additional 1 mo, and noncold-acclimated and cold-acclimated GL-3 and *MdHYL1* OE plants were exposed to −10 °C for 20 min and −10 °C for 25 min, respectively. After freezing treatment, all plants were transferred to a growth chamber for recovery. Bars = 3 cm. **B** and **D)** Survival rate of the plants shown in A and C. **E)** Ion leakage of noncold-acclimated and cold-acclimated GL-3 and *MdHYL1* OE plants. One-month-old plants were transferred to soil for an additional 1 mo, and leaves were collected for an ion leakage assay. **F)** Ion leakage from dormant branches. Error bars indicate standard deviation ($n = 12$–15 in **B** and **D** and 6 in **E** and **F**). The letters indicate significant differences according to 1-way ANOVA (Tukey's test; $P < 0.05$).

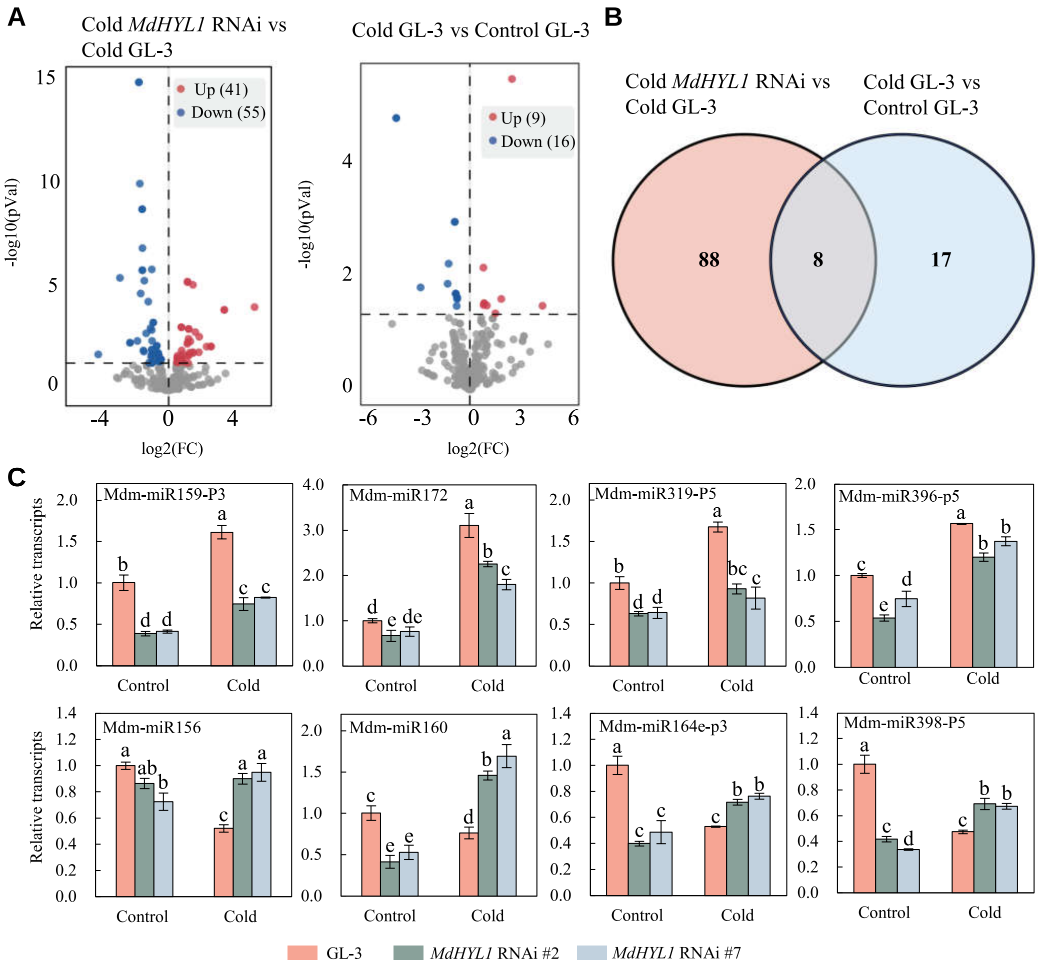

Figure 4. Small RNA-seq analysis of *MdHYL1* RNAi and GL-3 plants under control and cold treatment. **A)** Volcano plot showing upregulated and downregulated miRNAs in GL-3 and *MdHYL1* RNAi plants under control and cold treatment. **B)** Venn diagram showing the number of the cold-responsive miRNAs in GL-3 and *MdHYL1* RNAi plants. **C)** The relative transcripts of cold-responsive miRNAs. One-month-old plants were treated at 4 °C for 8 h, and leaves were collected for stem-loop RT-qPCR analysis. Error bars indicate standard deviation ($n = 3$ in **C**). The letters indicate significant differences according to 1-way ANOVA (Tukey's test; $P < 0.05$).

response to cold stress, transcription of Mdm-miR156 was repressed (Fig. 5A), while transcription of Mdm-miR172 was increased (Fig. 5B); this finding suggests opposite potential roles in cold response. To further understand the biological functions of Mdm-miR156 and Mdm-miR172, we generated transgenic calli that overexpressed Mdm-miR156 or Mdm-miR172 (Fig. 5, C and D). We then subjected these transgenic calli to a cold treatment for 15 d. After cold treatment, the relative growth rates of Mdm-miR156 OE lines were lower than that of the wild type (Fig. 5, E and F). In addition, we detected higher levels of H_2O_2 in Mdm-miR156 OE lines under cold; this finding was consistent with their repressed POD and CAT activities (Supplemental Fig. S8, A to C). Thus, these results indicate that Mdm-miR156 plays a negative role in apple cold tolerance. However, we found that the Mdm-miR172 OE lines showed a higher relative growth rate (Fig. 5, G and H), lower H_2O_2 content, and increased POD and CAT activities (Supplemental Fig. S8, D to F), suggesting that OE Mdm-miR172 can enhance cold tolerance in apple.

MdHYL1 enhances the expression of *MdMYB88* and *MdMYB124* in response to infection by *A. alternata* f. sp. *mali*

Since MdMYB88 and MdMYB124 also positively regulate disease resistance (Geng et al. 2020), we then wondered if MdHYL1 may exert similar pathogen response to cold stress effects. We first assessed the expression levels of *MdHYL1* in GL-3 plants treated with *A. alternata*. Our results showed that *MdHYL1* was significantly induced <24 h after infection (Supplemental Fig. S9A), which suggests that MdHYL1 may be involved in the response to *A. alternata* in apple. To examine if MdHYL1 positively regulated the expression of *MdMYB88* and *MdMYB124* in response to infection by *A. alternata*, we performed RT-qPCR analysis. As expected, we found repression of both *MdMYB88* and *MdMYB124* in *MdHYL1* RNAi transgenic plants but increased expression of both genes in *MdHYL1* OE transgenic plants relative to GL-3 control plants (Supplemental Fig. S9, B to E), which

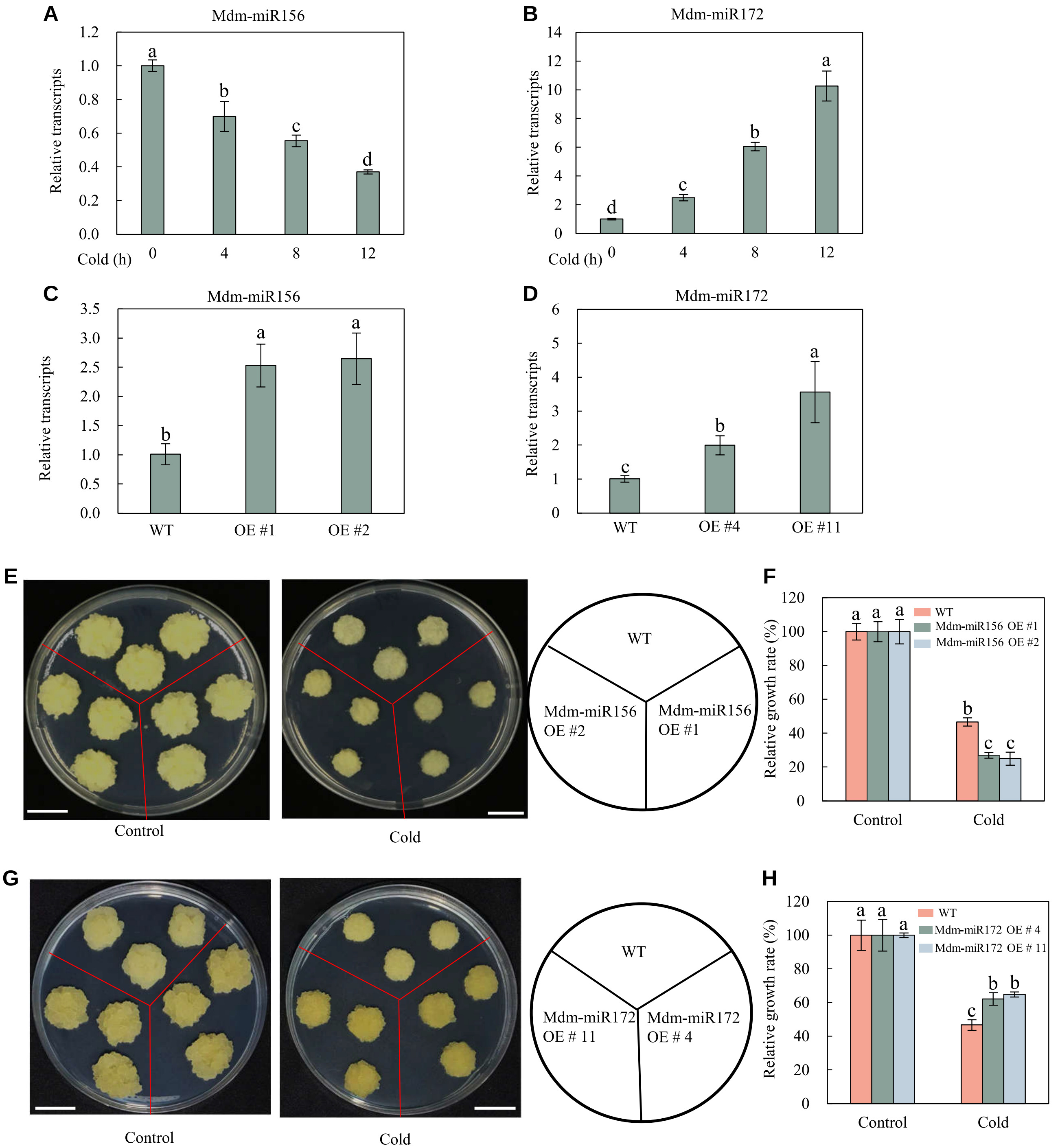

Figure 5. Cold tolerance of Mdm-miR156 and Mdm-miR172 overexpression (OE) calli. **A** and **B)** Expression pattern of Mdm-miR156 and Mdm-miR172 in response to cold stress. *M. domestica* plants were treated at 4 °C for 0, 4, 8, and 12 h, and leaves were collected for RNA extraction and RT-qPCR analysis. **C** and **D)** Relative transcripts of Mdm-miR156 in Mdm-miR156 OE calli and Mdm-miR172 in Mdm-miR172 OE calli. **E** and **F)** The relative growth rate of Mdm-miR156 OE calli under control and cold treatment. Bars = 1 cm. **G** and **H)** The relative growth rate of Mdm-miR172 OE calli under control and cold treatment. Equal weight wild-type and transgenic calli were grown on MS medium under control conditions for 7 d and then transferred to 4 °C for 15 d in the dark, followed by a calculation of relative growth rate. Bars = 1 cm. Error bars indicate standard deviation ($n = 3$ in **A** to **D** and 15 in **F** and **H**). The letters indicate significant differences according to 1-way ANOVA (Tukey's test; $P < 0.05$).

indicates that MdHYL1 positively regulates the expression of *MdMYB88* and *MdMYB124* in response to infection by *A. alternata*. We also detected the protein level of MdMYB88 and MdMYB124 in *MdHYL1* RNAi transgenic plants in response to *A. alternata* treatment. The results showed that MdHYL1 positively regulated the protein accumulation of

MdMYB88 and MdMYB124 under *A. alternata* treatment (Supplemental Fig. S9F). Similar to the regulatory pattern under cold, the expression of *MdHYL1* was not affected in both *MdMYB88* and *MdMYB124* OE plants and *MdMYB88/124* RNAi plants compared to GL-3 under *A. alternata* treatment (Supplemental Fig. S9G). Taken together, these results suggest that MdHYL1 positively regulates the gene expression and protein accumulation of MdMYB88 and MdMYB124 in response to infection of *A. alternata*.

MdHYL1 positively regulates *A. alternata* resistance in apple

We then attempted to determine the biological function that MdHYL1 plays in the infection response by infecting detached leaves and whole plants with *A. alternata*. After infection, we found that both detached leaves and whole plants of *MdHYL1* RNAi transgenic lines showed a greater inoculation area than GL-3 plant (Fig. 6, A to D). On the contrary, we found that detached leaves from *MdHYL1* OE transgenic plants had smaller disease spots than GL-3 plants (Fig. 6, E and G). Finally, inoculation of leaves from *MdHYL1* OE whole plants showed similar results (Fig. 6, F and H). Taken together, these data indicate that MdHYL1 plays a positive role in *A. alternata* infection resistance in apple.

Since MdHYL1 could regulate the expression of *MdMYB88* and *MdMYB124* in response to infection by *A. alternata* and MdMYB88 and MdMYB124 enhance pathogen resistance by inducing expression of *CHORISMATE MUTASE 2* (*CM2*) and phenylpropanoid content (Geng et al. 2020), we then examined the expression of *MdCM2* and plant phenylpropanoid content in *MdHYL1* OE and RNAi transgenic plants under control and *A. alternata* infection conditions. Our results showed that, after infection by *A. alternata*, both *MdCM2* expression and phenylpropanoid contents were higher in *MdHYL1* OE plants and lower in *MdHYL1* RNAi plants relative to GL-3 plants (Fig. 7, A to D). These results suggest that MdHYL1 positively regulates *A. alternata* resistance via regulating *MdCM2* expression and phenylpropanoid content in response to *A. alternata* infection.

MdHYL1 acts as upstream of *MdMYB88* during the *A. alternata* infection response

To further explore the genetic relationship between *MdHYL1* and *MdMYB88* in response to *A. alternata* infection, we transiently overexpressed *MdMYB88* in *MdHYL1* RNAi transgenic plant leaves (*MdMYB88* OE/*MdHYL1* RNAi) (Fig. 8A). After infection by *A. alternata* for 3 d, repression of *MdHYL1* decreased *A. alternata* resistance, while OE of *MdMYB88* enhanced *A. alternata* resistance in GL-3 plants (Fig. 8B). In addition, OE of *MdMYB88* and knockdown of *MdHYL1* expression partially recovered plant sensitivity to *A. alternata*, as compared with *MdHYL1* RNAi transgenic plants (Fig. 8B). Taken together, these data suggest that MdMYB88 is epistatic to MdHYL1 in the response of apple plants to *A. alternata* infection.

Next, to explore whether MdHYL1 regulates the production of phenylpropanoids through *MdMYB88* as part of the response to *A. alternata* infection, we measured phenylpropanoid content in *MdHYL1* OE, *MdHYL1* RNAi, and GL-3 plants in control and *A. alternata*-infected plants using UPLC-MS (ultra-performance liquid chromatography tandem mass spectrometry) technology. Our results showed that phenylpropanoid content decreased in plants with reduced *MdHYL1*, while OE of *MdMYB88* increased phenylpropanoid content relative to GL-3 control plants after infection by *A. alternata* (Fig. 8C). In addition, OE of *MdMYB88* and knockdown of *MdHYL1* partially rescued the reduced phenylpropanoid content phenotype caused by repression of *MdHYL1* (Fig. 8C). These data therefore suggest that *MdMYB88* acts downstream of MdHYL1 in the apple response to infection by *A. alternata*.

MdHYL1 regulates *A. alternata* resistance by modulating miRNA biogenesis

We next explored whether MdHYL1 affected the biogenesis of miRNAs in response to infection by *A. alternata*. sRNA-seq results showed that biogenesis of 102 Mdm-miRNAs was substantially affected after *A. alternata* infection, thereby indicating that apple may respond to *A. alternata* infection by mediating the biogenesis of Mdm-miRNAs. In *MdHYL1* RNAi plants, we found a substantially difference in the accumulation of 99 Mdm-miRNAs in response to *A. alternata* infection relative to GL-3 plants. Moreover, among these 99 Mdm-miRNAs, we identified 44 involved in plant response to infection by *A. alternata.* In addition, the repression of *MdHYL1* was associated with increased the accumulation of 62 Mdm-miRNAs and decreased the accumulation of 37 Mdm-miRNAs. These findings indicate that MdHYL1 regulates the biogenesis of Mdm-miRNAs in response to *A. alternata* infection (Fig. 9, A and B).

We selected 4 miRNAs (Mdm-miR156, Mdm-miR172, Mdm-miR164e-p3, and Mdm-miR160) to verify the sRNA-seq data via stem-loop RT-qPCR analysis. These 4 miRNAs were previously reported to participate in pathogen resistance. Silencing of miR156 confers enhanced resistance of the blast disease (*M. oryzae*) in rice (Zhang et al. 2020), and *mir156* mutant and *SPL9* OE plants display elevated sensitivity to *Botrytis cinerea* in *Arabidopsis* (Sun et al. 2022). OE of miR172a and b increases resistance to *Phytophthora infestans* infection in tomato (*Solanum lycopersicum*) (Luan et al. 2018). Knockdown of ghr-miR164 target, *GhNAC100*, increases cotton (*Gossypium hirsutum*) resistance to *Verticillium dahliae* (Hu et al. 2020). Interestingly, due to the fine-tuning regulatory manners of miRNAs, both OE and knockdown of miR160 result in enhanced potato (*Solanum chacoense*) susceptibility to *P. infestans* infection (Natarajan et al. 2018). Stem-loop RT-qPCR analysis revealed that the biogenesis of all the 4 Mdm-miRNAs was reduced in *MdHYL1* RNAi transgenic plants under control condition, as compared to GL-3. After infection by *A. alternata*,

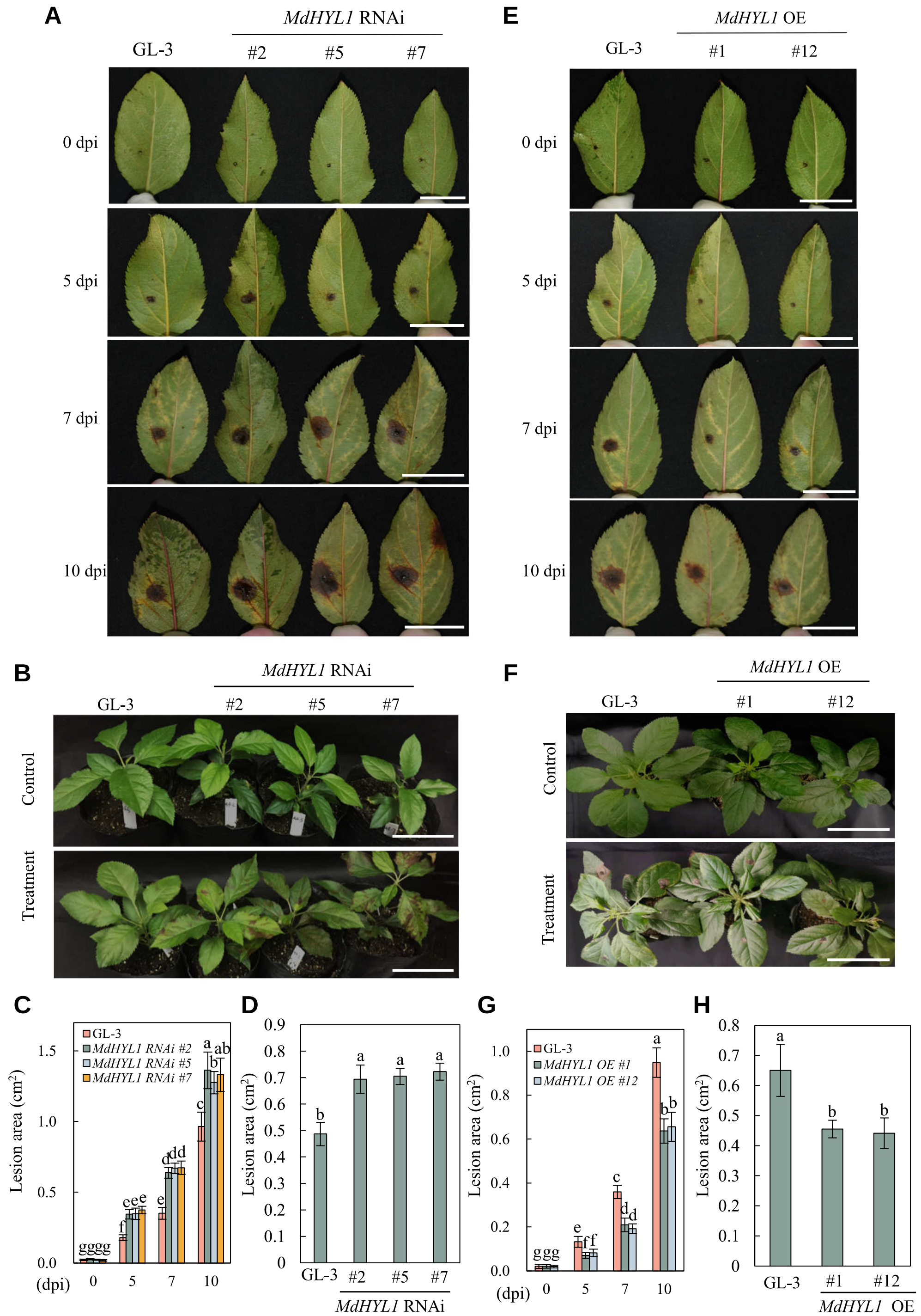

Figure 6. MdHYL1 positively regulates resistance to infection by *A. alternata* f. sp. *mali*. **A** and **B)** Disease resistance of detached leaves **(A)** and whole plants **(B)** of GL-3 and *MdHYL1* RNAi transgenic plants. dpi, days post-infection. Bars = 3 cm. **C** and **D)** Quantification of the data shown in **A** and **B**. **E** and **F)** Disease resistance of detached leaves **(E)** and whole plants **(F)** of GL-3 and *MdHYL1* OE plants. dpi, days post-infection. Bars = 4 cm. **G** and **H)** Quantification of the data shown in **E** and **F**. OE, overexpression. Error bars indicate standard deviation ($n = 15$ in **C**, **D**, **G**, and **H**). The letters indicate significant differences according to 1-way ANOVA (Tukey's test; $P < 0.05$).

except for Mdm-miR172, transcripts of Mdm-miR156, Mdm-miR164e-p3, and Mdm-miR160 were reduced in *MdHYL1* RNAi plants compared to GL-3. Consistently, transcripts of these 4 Mdm-miRNAs were induced by *A. alternata* except for Mdm-miR172. These results suggest the potential involvement of these Mdm-miRNAs in MdHYL1-mediated pathogen resistance (Fig. 9C). To confirm that these Mdm-miRNAs contributed to *A. alternata* resistance in

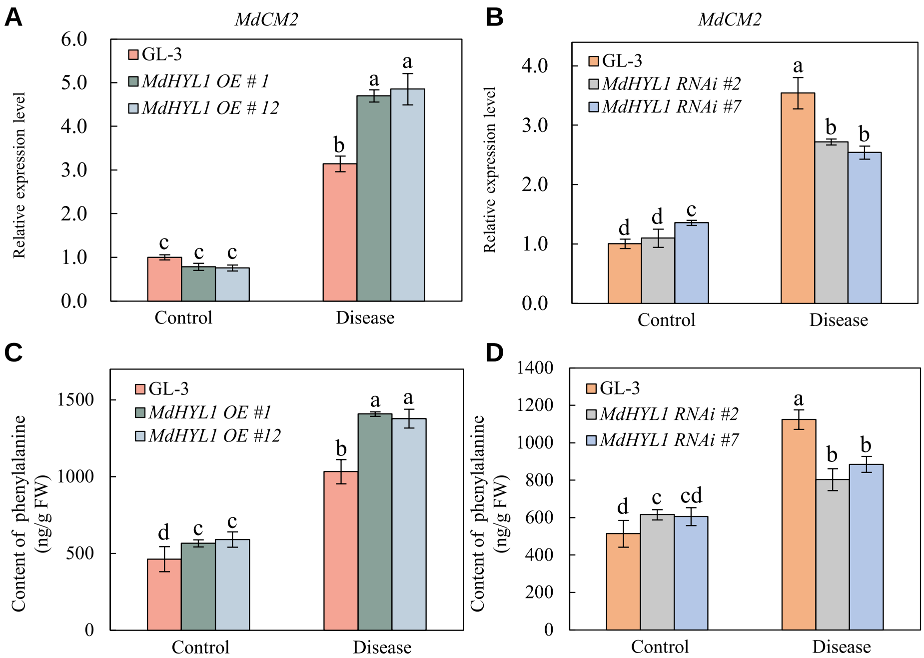

Figure 7. The expression level of *MdCM2* and phenylalanine content of GL-3 and *MdHYL1* transgenic plants. **A** and **B)** The expression level of *MdCM2* in *MdHYL1* OE and RNAi plants under control and *A. alternata* f. sp. *mali* treatment. One-month-old transgenic plant leaves were treated with *A. alternata* for 24 h, and the infected leaves were collected for RT-qPCR analysis. **C** and **D)** The phenylalanine content in GL-3 and *MdHYL1* transgenic plants. One-month-old transgenic plant leaves were treated by *A. alternata* for 3 d, and the infected leaves were collected for detection of phenylalanine content by UPLC-MS analysis. OE, overexpression. Error bars indicate standard deviation ($n = 3$ in **A** and **B** and 9 in **C** and **D**). The letters indicate significant differences according to 1-way ANOVA (Tukey's test; $P < 0.05$).

apple, we evaluated the resistance of Mdm-miR160e OE plants generated by a previous study (Shen et al. 2022) to infection by *A. alternata*. We found that Mdm-miR160e OE plants showed greater lesion areas than GL-3 control plants after infection by *A. alternata* for 5, 7, and 10 d (Fig. 9, D and E), and these findings suggest that Mdm-miR160 is a negative regulator of *A. alternata* resistance in apple.

Discussion

HYL1 has been widely found to participate in plant development. In *Arabidopsis*, a *hyl1* mutant was first characterized as having shorter plant height, leaf hyponasty, and decreased root growth rate (Lu and Fedoroff 2000). Despite the crucial role it plays in plant development, little is known regarding the function played by HYL1 in response to biotic and abiotic stress. It has been reported that the *hyl1* mutant is hypersensitive to ABA but is less sensitive to cytokinin and auxin (Lu and Fedoroff 2000). In addition, the *hyl1* mutant has also been found to be more sensitive to tunicamycin, a compound causing endoplasmic reticulum stress relative to wild-type plants (Hirata et al. 2019). In apple, MdHYL1 has been shown to regulate miRNA biogenesis and drought tolerance (You et al. 2014; Shen et al. 2022). In this study, we evaluated the cold tolerance and pathogen resistance of *MdHYL1* transgenic apple plants. We found that *MdHYL1* RNAi plants were more sensitive to cold stress (Fig. 2) and *MdHYL1* OE plants were more tolerant to freezing stress (Fig. 3). In addition, *MdHYL1* OE plants were more tolerant while *MdHYL1* RNAi plants were more sensitive to the infection by *A. alternata* than GL-3 plants (Fig. 6). These results indicate that MdHYL1 plays a positive role in the apple cold tolerance and pathogen resistance.

In plants, MYB transcription factors are one of the largest protein families. They are involved in diverse processes, including plant development and responses to biotic and abiotic stress (Baillo et al. 2019; Wang et al. 2021). Among apple MYB proteins, MdMYB88 and MdMYB124 have been reported to participate in plant development as well as both biotic and abiotic stress. OE of *MdMYB88* and *MdMYB124* has been found to facilitate root xylem development and ABA accumulation in apple trees, thereby improving drought tolerance (Geng et al. 2018; Xie et al. 2021). *MdMYB88* and *MdMYB124* are also induced by cold stress and positively regulate freezing tolerance through both CBF-dependent and CBF-independent pathways (Xie et al. 2018). Moreover, MdMYB88 can bind to the promoter region of *MdCM2* to enhance phenylalanine biosynthesis, which increases drought tolerance and disease resistance in apple trees (Geng et al. 2020). Previous research has also shown that MdMYB88 and MdMYB124 can interact with MdHYL1 (Niu et al. 2022). In the current study, we showed that MdHYL1 positively regulated the expression of *MdMYB88* and *MdMYB124* under cold and *A. alternata* treatment, but not

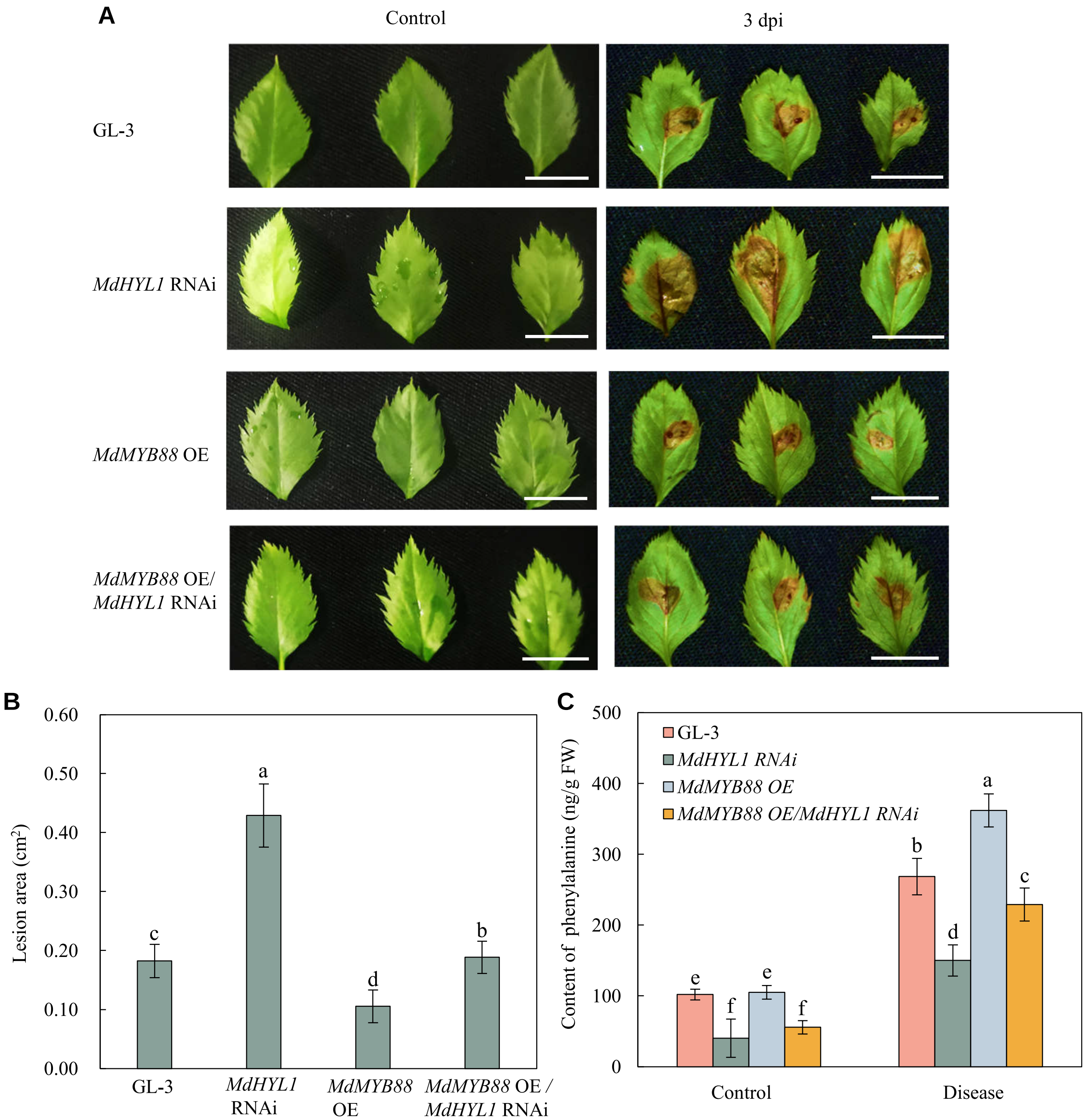

Figure 8. MdHYL1 positively regulates resistance to *A. alternata* f. sp. *mali* infection via MdMYB88. **A)** Disease symptoms of GL-3, *MdHYL1* RNAi, *MdMYB88* OE, and *MdMYB88* OE/*MdHYL1* RNAi transgenic leaves. *MdMYB88* OE/*MdHYL1* RNAi was obtained by transiently overexpressing *MdMYB88* in *MdHYL1* RNAi plants. dpi, days post-infection. Bars = 1 cm. **B)** Quantification of the data shown in **A**. **C)** The phenylalanine content in GL-3, *MdHYL1* RNAi, *MdMYB88* OE, and *MdMYB88* OE/*MdHYL1* RNAi transgenic plants. OE, overexpression. One-month-old transgenic plant leaves were treated with *A. alternata* for 3 d, and the infected leaves were collected for detection of phenylalanine content by UPLC-MS analysis. Error bars indicate standard deviation ($n = 15$ in **B** and 9 in **C**). The letters indicate significant differences according to 1-way ANOVA (Tukey's test; $P < 0.05$).

vice versa (Supplemental Figs. S1 and S9). Moreover, *MdHYL1* played a positive role in apple cold tolerance and *A. alternata* resistance by mediating ROS scavenging, anthocyanin accumulation, and phenylalanine content (Figs. 2, 3, and 8; Supplemental Figs. S3 to S5), which incidentally showed consistency with MdMYB88 and MdMYB124 under cold stress and pathogen infection. Furthermore, we also found that OE of *MdMYB88* increased the cold tolerance of *MdHYL1* RNAi calli (Supplemental Fig. S6), as well as increased the pathogen resistance of *MdHYL1* RNAi transgenic leaves (Fig. 8). These results suggested that MdHYL1 acts upstream of MdMYB88 with respect to both cold tolerance and pathogen resistance.

HYL1 is a critical factor in miRNA biogenesis in many plant species (D'Ario et al. 2017; Shen et al. 2022). In the current study, sRNA-seq analysis identified 96 miRNAs whose biogenesis was affected in *MdHYL1* RNAi plants under cold, including Mdm-miR396-p5, Mdm-miR160, Mdm-miR319-p5,

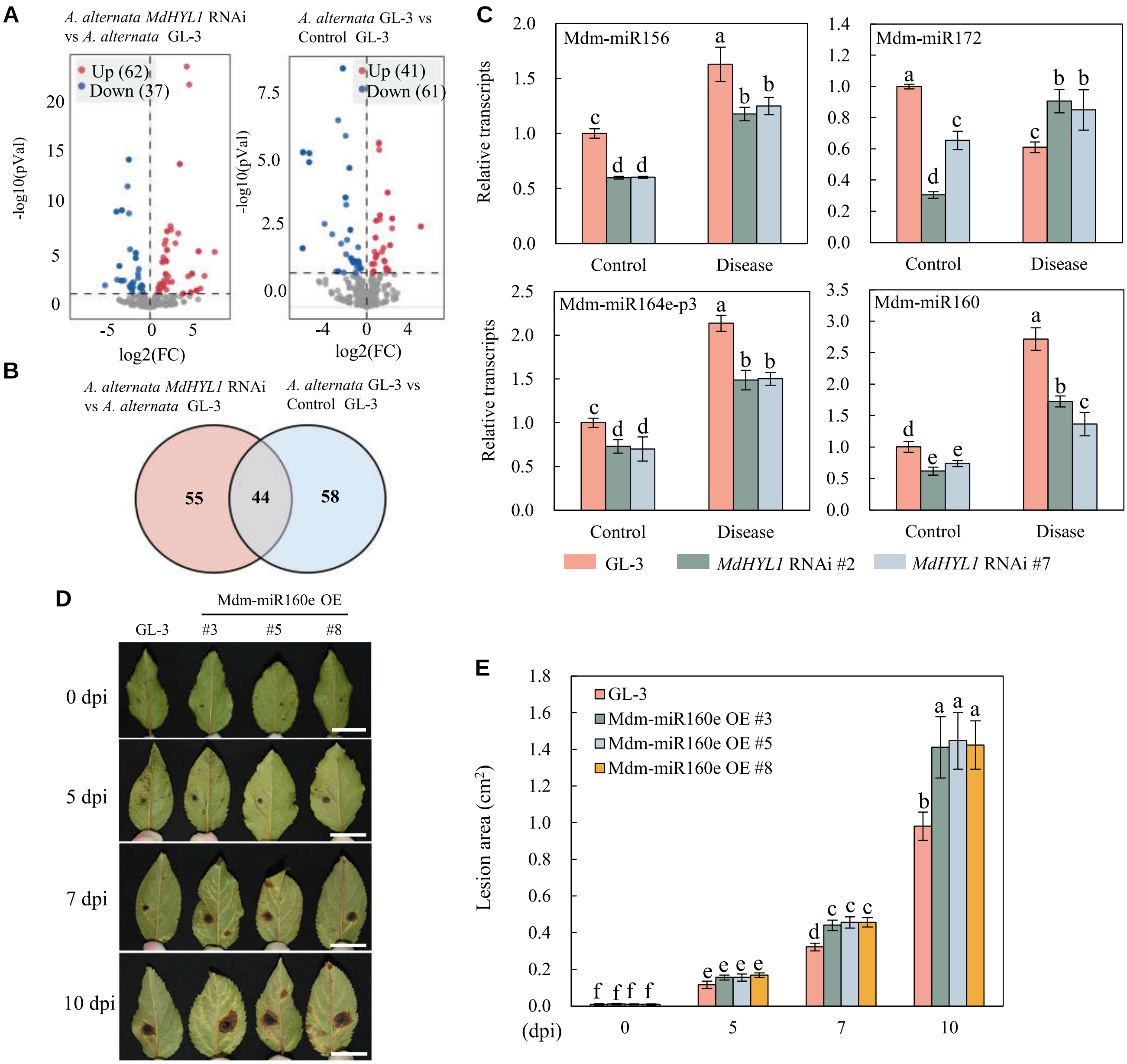

Figure 9. Small RNA-seq analysis of *MdHYL1* RNAi and GL-3 plants under control and *A. alternata* f. sp. *mali* treatment. **A)** Volcano plot showing upregulated and downregulated miRNAs in GL-3 and *MdHYL1* RNAi plants under control and disease treatment. **B)** Venn diagram showing the number of the infection-responsive miRNAs in GL-3 and *MdHYL1* RNAi plants. **C)** The relative transcripts of disease-responsive miRNAs in GL-3 and *MdHYL1* RNAi plants under control and disease treatment. One-month-old plants were infected with *A. alternata* f. sp. *mali* for 24 h, and leaves were collected for stem-loop RT-qPCR analysis. **D)** Disease resistance of GL-3 and Mdm-miR160e OE detached leaves. dpi, days post-infection. Bars = 3 cm. **E)** Quantification of the data shown in **D**. OE, overexpression. Error bars indicate standard deviation ($n = 3$ in **C** and 15 in **E**). The letters indicate significant differences according to 1-way ANOVA (Tukey's test; $P < 0.05$).

Mdm-miR156, Mdm-miR172, Mdm-miR164e-p3, Mdm-miR159-p3, and Mdm-miR398-p5. Using transgenic apple calli, we found that Mdm-miR156 was a negative while Mdm-miR172 was a positive regulator for apple cold tolerance, which were consistent with the function of their homologs in rice and wheat (*Triticum aestivum*) (Gupta et al. 2014; Zhou and Tang 2019). These results suggest that MdHYL1 may improve cold tolerance by regulating the biogenesis of cold-responsive miRNAs including Mdm-miR156 and Mdm-miR172. In addition, the biogenesis of 99 pathogen-responsive miRNAs was also affected by the reduction of *MdHYL1* (Fig. 9B), indicating that MdHYL1 may positively regulate pathogen resistance by affecting the biogenesis of these pathogen-responsive miRNAs, including Mdm-miR160 which was a negative regulator in apple *A. alternata* resistance. Furthermore, we also found that Mdm-miR156, as well as Mdm-miR160, was repressed in response to cold stress, while it was induced in response to infection by *A.*

alternata. In addition, the accumulation of Mdm-miR172 was induced in response to cold but was repressed by exposure to *A. alternata* infection. These results illustrate the complicated regulation mechanisms of miRNAs in response to distinct biotic and abiotic stresses.

Previous studies have revealed a tight relationship between cold response and disease response in plant. For example, the expression of many defense-related genes is induced in *Vitis amurensis* under cold, including *JAZ1* and *WRKY30* (Wu et al. 2014). In triticale (*Triticosecale* Wittm.), cold acclimation enhanced the plant genotype-dependent resistance against *Microdochium nivale* (Szechyńska-Hebda et al. 2015). Besides, many genes are induced in both cold and pathogen stresses, such as protein phosphatase 2C (Hu et al. 2010; Widjaja et al. 2010), stilbene synthase (Xu et al. 2010; Dai et al. 2012), and 9-*cis*-epoxycarotenoid dioxygenase (Fan et al. 2009; Xian et al. 2014). In this study, we determined that MdHYL1 positively regulated cold tolerance and pathogen resistance via modulation of MdMYB88 and MdMYB124 transcription and biogenesis of Mdm-miRNAs. Since MdMYB88 and MdMYB124 have been proved to participate in cold and disease responses in apple (Geng et al. 2018; Xie et al. 2018; Geng et al. 2020), we consider both MdMYB88 and MdMYB124 might act as the key molecular switch regulating cold stress response and pathogen defense in apple, which may benefit genetic engineering of plant breeding in the future.

Materials and methods

Plant materials, growth conditions, and stress treatment

To analyze the gene expression pattern in response to cold stress, 'Golden Delicious' apple (*M. domestica*) trees grown in the greenhouse were used for RNA extraction. One-month-old tissue-cultured GL-3 (Dai et al. 2013) and transgenic plants were rooted in 1/2 MS medium (2.215 g/L MS salts, 20 g/L sucrose, and 7.5 g/L agar, pH 5.8) supplemented with 0.5 mg/L indole-3-butytric acid and 0.5 mg/L indole-acetic acid under dark conditions for 3 d and then held under long-day conditions (14-h light/10-h dark) for additional 2 mo. Plants were then transplanted into soil and grown in a light growth chamber with 60% humidity under long-day conditions. 'Orin' (*M. domestica*) calli were grown on MS media supplemented with 0.4 mg/L 6-BA and 1.5/mg L 2,4-dichlorophenoxyacetic acid (2,4-D) in the dark.

Briefly, to test the accumulation of ROS and the activity of antioxidant enzymes in GL-3 and *MdHYL1* transgenic plants after cold treatment, plants were treated by being exposed to 4 °C for 8 h. Moreover, plants were exposed to 4 °C for 7 d for cold acclimation. Freezing treatment was performed as described previously (Xie et al. 2018). To examine the survival rate, the noncold-acclimated plants were placed at −10 °C for 20 min in the low-temperature chamber (Thermo Fisher, BPHJS-120B), while the cold-acclimated plants were exposed to −10 °C for 25 min. After treatment, all the plants were recovered at normal conditions. Two weeks later, the plants which had new leaves in internodes or apical meristem were counted. Fifteen apple seedlings were used for cold treatment and further survival rate analyses. The experiments were repeated 3 times, and representative data from 1 experiment were shown in the Figure.

Cold tolerance of the transgenic calli and the wild type was evaluated as previously described with minor modifications (An et al. 2020). To examine the relative growth rates of calli under cold treatment, equal weight of the wild-type and transgenic calli was grown on the same plate, cultured at normal temperature for 7 d, and then treated at 4 °C for 15 d in the dark, followed by a calculation of relative growth rate.

Vector construction and calli transformation

For the *MdMYB88* OE construct, coding regions of the *MdMYB88* were cloned into a pCAMBIA1305 vector. *MdHYL1* RNAi and OE constructs were obtained previously (Shen et al. 2022). For the OE constructs of Mdm-miR156 and Mdm-miR172, the primary sequence of 384-bp Mdm-miR156a and 368-bp Mdm-miR172c was cloned into the vector pDONR222 and then to the destination vector pK2GW7 by the Gateway technology (Invitrogen, Thermo Scientific, USA). The transformation method of apple calli was described previously (An et al. 2021); *MdHYL1* RNAi and *MdMYB88* OE vectors for apple calli transformation were generated before.

The primers used are listed in Supplemental Table S1.

Determination of ion leakage

Ion leakage of leaves was detected as described previously (Guan et al. 2013; Xie et al. 2018). Leaves of noncold-acclimated and cold-acclimated GL-3 and *MdHYL1* OE transgenic plants were punched into the 50-mm diameter discs and then placed into a test tube containing 1 ml of deionized H_2O and a 1-ml ice cube. The tubes were then placed into the low-temperature aqua bath cycle instrument (Thermo Fisher PC200-A40), and the freezing temperature gradient (0 °C, −4 °C, and −6 °C) was set as previously described (Xie et al. 2018). Six-leaf discs in a tube were used as 1 biological replicate, and each treatment had 6 tubes.

Ion leakage of dormant branches under freezing stress was performed as described previously (Schreiber et al. 2013). All samples were taken from 1-yr-old dormant branches, which were cut into 2-mm pieces and placed into a test tube containing 1-ml deionized H_2O. Six-branch cuts in a tube were used as 1 biological replicate, and each treatment had 6 tubes.

Small RNA-sequencing

One-month-old GL-3 and *MdHYL1* RNAi transgenic apple plants were exposed to 4°C for 8 h, and then the third to fifth leaves were collected for small RNA-sequencing. Three plants were used as a biological replicate, and each line had 3 biological replicates. For disease treatment, the detached leaves of GL-3 and *MdHYL1* RNAi transgenic plants were infected by *A. alternata* for 3 d, and the infected leaves were collected for sRNA-seq. Six leaves were used as a biological replicate, and each treatment had 3 biological replicates.

High-quality total RNA was extracted from approximately 100 mg of the collected treated leaves using CTAB methods, and about 10 mg of enriched small RNA was then isolated by Illumina TruSeq Small RNA Sample Prepkit (Illumina, San Diego, CA, USA) and used for small RNA library construction. Small RNA library quality assessment was performed on an Agilent 2100 Bioanalyzer (Agilent, Palo Alto, CA, USA) and a Step-One Plus Real-Time PCR system (ABI, Carlsbad, CA, USA). Deep sequencing was performed on an Illumina HiSeq 2000 system (Illumina).

RT-qPCR analysis and stem-loop Rt-qPCR analysis

RT-qPCR analysis was carried out according to the previous description (Xie et al. 2018). Stem-loop RT-qPCR analysis was performed as described (Niu et al. 2019). Briefly, high-quality total RNA was reversed by the RevertAid First Strand cDNA Synthesis Kit (Thermo Scientific, K1622, USA). For stem-loop RT-qPCR, the reverse transcription primers contained miRNA-specific reverse transcription primer and the *malate dehydrogenase* (MDH) reverse primer (Varkonyi-Gasic et al. 2007). Three biological replicates were used, and each replicate contained leaves from 3 plants. MDH was used as reference gene, and the relative expression was calculated using $2^{-\Delta\Delta Ct}$ method (Livak and Schmittgen 2001).

The primers are listed in Supplemental Table S1.

Determination of POD and CAT activities and the content of H_2O_2

Content of H_2O_2 was measured by a H_2O_2 assay kit (#BC3590, Solarbio, Beijing, China); POD and CAT activities were determined by the previous methods (Liu et al. 2014). The GL-3 and transgenic plants were treated at 4 °C for 8 h, and then leaves were collected for determination.

Disease resistance analysis

Disease resistance analysis of detached leaves and whole plants of GL-3 and *MdHYL1* RNAi and OE transgenic plants was performed according to a previous report (Li et al. 2019). The representative photographs of detached leaves were taken at 0, 5, 7, and 10 d post-infection (dpi), and the pictures of whole plants were taken at 10 dpi, and then the lesion area was calculated by ImageJ software. The transiently transgenic leaves were also used for *A. alternata* resistance analysis as described previously (Liu et al. 2021). *MdMYB88* OE/*MdHYL1* RNAi double genes were performed by transiently transforming the *MdMYB88* OE vector into the *MdHYL1* RNAi leaves for 3 d for *A. alternata* incubation. Three biological replicates were applied, and 15 leaves were used as 1 replicate. This experiment was repeated 3 times.

Measurement of phenylalanine content

For measurement of the phenylalanine content, the leaves of the GL-3, *MdHYL1* OE, and RNAi plants under both control and *A. alternata* treatment were collected for UPLC-MS analysis.

UPLC-MS analysis was performed according to the previous reports with modifications (Chen et al. 2013; Geng et al. 2020). Freeze-dried leaves were ground into fine powder using liquid nitrogen, and 100 mg of powder was then weighed and extracted overnight with 1 ml of 70% aqueous methanol at 4 °C. Following centrifugation at 4 °C for 10 min (12,000 × *g*), the extracts were transferred to another 1-ml centrifuge tube and then filtered with a filter membrane (0.22 μm). The extracts were then analyzed using a Liquid Chromatography Electrospray Ionization Tandem Mass Spectrometry (LC-ESI-MS/MS) system (Applied Biosystems 6500 QTRAP, USA).

The analytical conditions were as follows: UPLC column, Waters ACQUITY UPLC HSS T3 C18 (1.8 μm, 2.1 mm × 100 mm); solvent system, water [0.04% (v/v) acetic acid]: acetonitrile [0.04% (v/v) acetic acid]; and gradient program, flow rate, and injection volume were the same as described previously (Geng et al. 2020). The effluent was alternatively connected to an ESI-triple quadrupole-linear ion trap mass spectrometer (QTRAP-MS). LIT and triple quadrupole scans were acquired by an API 6500 QTRAP LC/MS/MS triple QTRAP-MS system. The ESI source operation parameters were the same as described previously (Chen et al. 2013).

Statistical analysis

Statistical analyses were reported as the mean ± SD and determined by 1-way ANOVA (Tukey's test; $P < 0.05$).

Accession numbers

Sequence data from this article can be found in the GenBank/EMBL data libraries under accession numbers: MdHYL1 (XP_028950144.1), MdMYB88 (KY569647), MdMYB124 (KY569648), MdCM2 (XP_008379007), MdCSP3 (XP_008340702.1), MdCBF1 (XP_008377096.1), MdCBF3 (NP_001315892.1), MdCOR47 (XP_008352066), MdCCA1 (XP_008343467.1), and MdSIZ1 (XP_028944339.1).

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Author Contributions

Q.G., X.L., and X.S. conceived and designed the project; X.S., Y.S., Y.P., Y.X., and X.L. performed experiments. J.H. analyzed data; Y.S., Q.G., and X.S. wrote the manuscript; F.M. helped with the discussion of the work. All the authors contributed to the preparation of the final manuscript. The author responsible for distribution of materials integral to the findings presented in this article in accordance with the policy described in the Instructions for Authors (https://academic.oup.com/plphys/pages/General-Instructions) is Qingmei Guan (qguan@nwafu.edu.cn).

Supplemental data

The following materials are available in the online version of this article.

Supplemental Figure S1. Regulation of *MdMYB88* and *MdMYB124* by MdHYL1 under control and cold treatment.

Supplemental Figure S2. MdCSP3 promotes the cold hardiness of apples.

Supplemental Figure S3. Antioxidant enzyme activities of *MdHYL1* RNAi plants under control and cold treatment.

Supplemental Figure S4. Antioxidant enzyme activities of *MdHYL1* OE plants under control and cold treatment.

Supplemental Figure S5. MdHYL1 promotes anthocyanin accumulation in apple under cold stress.

Supplemental Figure S6. MdMYB88 acts downstream of MdHYL1 in response to cold stress.

Supplemental Figure S7. Pearson correlation analysis of GL-3 and *MdHYL1* RNAi plants under control, cold, and *A. alternata* treatment.

Supplemental Figure S8. Antioxidant enzyme activities of Mdm-miR156 and Mdm-miR172 OE calli.

Supplemental Figure S9. Regulation of *MdMYB88* and *MdMYB124* by MdHYL1 in response to infection by *A. alternata* f. sp. *mali*.

Supplemental Table S1. Primers used in this study

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Conflict of interest statement. The authors declare that they have no conflicts of interest.

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